

Testing Report NutriGuide

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Repository:

Frontend: <https://github.com/Adisonsmn/frontend-nutriguide>

Backend: <https://github.com/Adisonsmn/backend-nutriguide>

Tanggal Pengujian: 2026-07-08

Tool Unit Test: Vitest v4.1.7

Fitur yang Diuji

No	Fitur	Modul/Service
1	Autentikasi (Register & Login)	auth.service.ts
2	Kalkulasi Nutrisi (BMR / TDEE)	nutrition.service.ts
3	Profile Management	profile.service.ts
4	History Makanan	history.service.ts
5	Lupa & Reset Password (OTP Flow)	auth.service.ts

Test Case

Fitur 1 Autentikasi (Register & Login)

No	Fitur	Skenario	Input	Expected Result	Status
1	Register	Email sudah terdaftar	email: "john@example.com" (sudah ada di DB)	Error 409 "Email already registered"	Pass
2	Register	Data valid, user baru	name: "John", email: "john@example.com", password: "pwd123"	Mengembalikan objek user (user_id, name, email), password di-hash dengan bcrypt	Pass
3	Register	Password di-hash dengan benar	password: "pwd123"	bcrypt.hash dipanggil dengan salt 10, hasil hash tersimpan di DB	Pass

No	Fitur	Skenario	Input	Expected Result	Status
4	Login	Email tidak terdaftar	email: "notfound@example.com"	Error 401 "Email not registered"	Pass
5	Login	Password salah	Email terdaftar + password: "salah"	Error 401 "Incorrect password"	Pass
6	Login	Kredensial benar	Email & password valid	Mengembalikan accessToken, refreshToken, dan data user; sesi tersimpan di DB	Pass

Fitur 2 Kalkulasi Nutrisi (BMR / TDEE)

No	Fitur	Skenario	Input	Expected Result	Status
7	Kalkulasi Nutrisi	Profil tidak ditemukan	userId: "USER-999"	Error 404 "Profile not found. Please create your profile first."	Pass
8	Kalkulasi Nutrisi	Pria, goal turun berat, aktivitas moderat	gender: Male, weight: 80kg, height: 180cm, age: 25, goal: lose_weight, activity: moderate	bmr: 1805, tdee: 2798, dailyCalorieTarget: 2298, macros: {protein:172, carbs:230, fat:77}	Pass
9	Kalkulasi Nutrisi	Wanita, goal naik berat, aktivitas sedentary	gender: Female, weight: 60kg, height: 165cm, age: 30, goal: gain_weight, activity: sedentary	bmr: 1320, tdee: 1584, dailyCalorieTarget: 2084, macros: {protein:156, carbs:208, fat:69}	Pass
10	Kalkulasi Nutrisi	Goal tidak dikenali (maintain weight)	goal: "maintain weight", aktivitas sedentary	dailyCalorieTarget === tdee (tidak ada penambahan/pengurangan kalori)	Pass

No	Fitur	Skenario	Input	Expected Result	Status
11	Info Nutrisi Makanan	Food ID tidak ditemukan	foodId: "FOOD-999"	Error 404 "Food not found"	Pass
12	Info Nutrisi Makanan	Food ID valid	foodId: "FOOD-001"	Mengembalikan data makanan lengkap beserta info resep	Pass

Fitur 3 Profile Management

No	Fitur	Skenario	Input	Expected Result	Status
13	Buat Profil	Profil sudah ada	userId: "USER-001" (sudah punya profil)	Error 409 — "Profile already exists"	Pass
14	Buat Profil	Data valid, profil baru	userId: "USER-002", age:25, weight_kg:70, height_cm:175, gender:"Male", goal:"lose_weight"	Profil berhasil dibuat, notifikasi terkirim ke user	Pass
15	Ambil Profil	Profil dan preferensi ada	userId: "USER-001"	Mengembalikan objek {profile, preferences} dengan data lengkap	Pass
16	Update Profil	Update sebagian field	userId: "USER-001", {weight_kg: 75}	Profil ter-update, notifikasi "Profile updated successfully" terkirim	Pass
17	Upsert Preferensi	Preferensi belum ada (create)	userId: "USER-001", diet_type:"vegetarian", daily_budget:50000	Preferensi baru dibuat dengan ID baru (PREF-xxx)	Pass

No	Fitur	Skenario	Input	Expected Result	Status
18	Upsert Preferensi	Preferensi sudah ada (update)	userId: "USER-001", daily_budget:70000	Preferensi ter-update tanpa membuat ID baru (Bug #19 fix)	Pass

Fitur 4 History Makanan

No	Fitur	Skenario	Input	Expected Result	Status
19	Tambah History	Food ID tidak ditemukan	foodId: "FOOD-999"	Error 404 — "Food not found"	Pass
20	Tambah History	Makanan disimpan (belum dikonsumsi)	userId:"USER-001", foodId:"FOOD-001", qty:150g, isConsumed:false	Entry tersimpan, notifikasi "Saved ... to your food history" terkirim	Pass
21	Tambah History	Makanan dikonsumsi	userId:"USER-001", foodId:"FOOD-002", qty:100g, isConsumed:true	Entry tersimpan dengan is_consumed:true, notifikasi "You consumed ... nutrition logged!"	Pass
22	Ambil History	Tanpa filter tanggal	userId:"USER-001"	Semua riwayat user dikembalikan, diurutkan descending by consumed_at	Pass
23	Ambil History	Dengan filter tanggal	userId:"USER-001", date:"2026-05-27"	Hanya riwayat tanggal 27 Mei 2026 yang dikembalikan (range 00:00 - 23:59)	Pass
24	Hapus History	Entry tidak ditemukan	historyId: "HIST-999"	Error 404 — "History entry not found"	Pass
25	Hapus History	Entry milik user lain (unauthorized)	userId:"USER-001", historyId	Error 403 — "You are not authorized to delete this entry"	Pass

No	Fitur	Skenario	Input	Expected Result	Status
			milik "USER-002"		
26	Hapus History	Entry valid milik user sendiri	userId:"USER-001", historyId yang benar	Entry berhasil dihapus dari database	Pass
27	Ringkasan Harian	Hitung total kalori & makro	Banana 150g + Chicken 200g, timezone WIB (+420 min)	totalCalories:464, remaining:1537, percentage:23, macros:{protein:64, carbs:35, fat:8}	Pass

Fitur 5 — Lupa Password & Reset Password (OTP Flow)

No	Fitur	Skenario	Input	Expected Result	Status
28	Refresh Token	JWT tidak valid / expired	refreshToken: "bad-token"	Error 401 — "Invalid or expired refresh token"	Pass
29	Refresh Token	Tidak ada sesi aktif di DB yang cocok	Token JWT valid tapi tidak ada hash yang cocok di DB	Error 401 — "Invalid refresh token"	Pass
30	Refresh Token	Token valid & sesi cocok	Token valid, sesi aktif, hash cocok	Mengembalikan accessToken baru	Pass
31	Lupa Password	Email tidak ditemukan	email: "nobody@example.com"	Error 404 — "Email not found"	Pass
32	Lupa Password	Email valid, OTP berhasil dikirim	email: "john@example.com"	OTP 6 digit tersimpan di DB + dikirim via email, response "OTP has been sent..."	Pass
33	Lupa Password	Pengiriman email gagal (SMTP error)	SMTP error saat kirim email	Error 500 — "Failed to send OTP email", OTP tidak	Pass

No	Fitur	Skenario	Input	Expected Result	Status
				dikembalikan ke client (Bug #1 fix)	
34	Reset Password	Email tidak ditemukan	email: "ghost@example.com"	Error 404 — "Email not found"	Pass
35	Reset Password	OTP salah	otp: "000000" (bukan OTP yang tersimpan)	Error 400 — "Invalid OTP"	Pass
36	Reset Password	OTP sudah kedaluwarsa	OTP benar tapi reset_otp_expires di masa lalu	Error 400 — "OTP has expired"	Pass
37	Reset Password	Password baru = password lama	bcrypt.compare mengembalikan true	Error 400 — "New password must be different from your current password"	Pass
38	Reset Password	Semua valid, reset berhasil	OTP benar + belum expired + password berbeda	Password hash ter-update, OTP di-null-kan, notifikasi terkirim, response "Password reset successfully"	Pass

Rekap Test Per File

File Test	Jumlah Test	Fitur
auth.service.test.ts	17	Fitur 1 (Register, Login, Logout) + Fitur 5 (Refresh Token, Forgot Password, Reset Password)
nutrition.service.test.ts	6	Fitur 2 (Kalkulasi BMR/TDEE + Info Makanan)
profile.service.test.ts	6	Fitur 3 (Create, Get, Update Profile, Upsert Preferences)
history.service.test.ts	9	Fitur 4 (Add, Get, Delete History, Daily Summary)

File Test	Jumlah Test	Fitur
Total	38	5 Fitur

Perhitungan Metrik

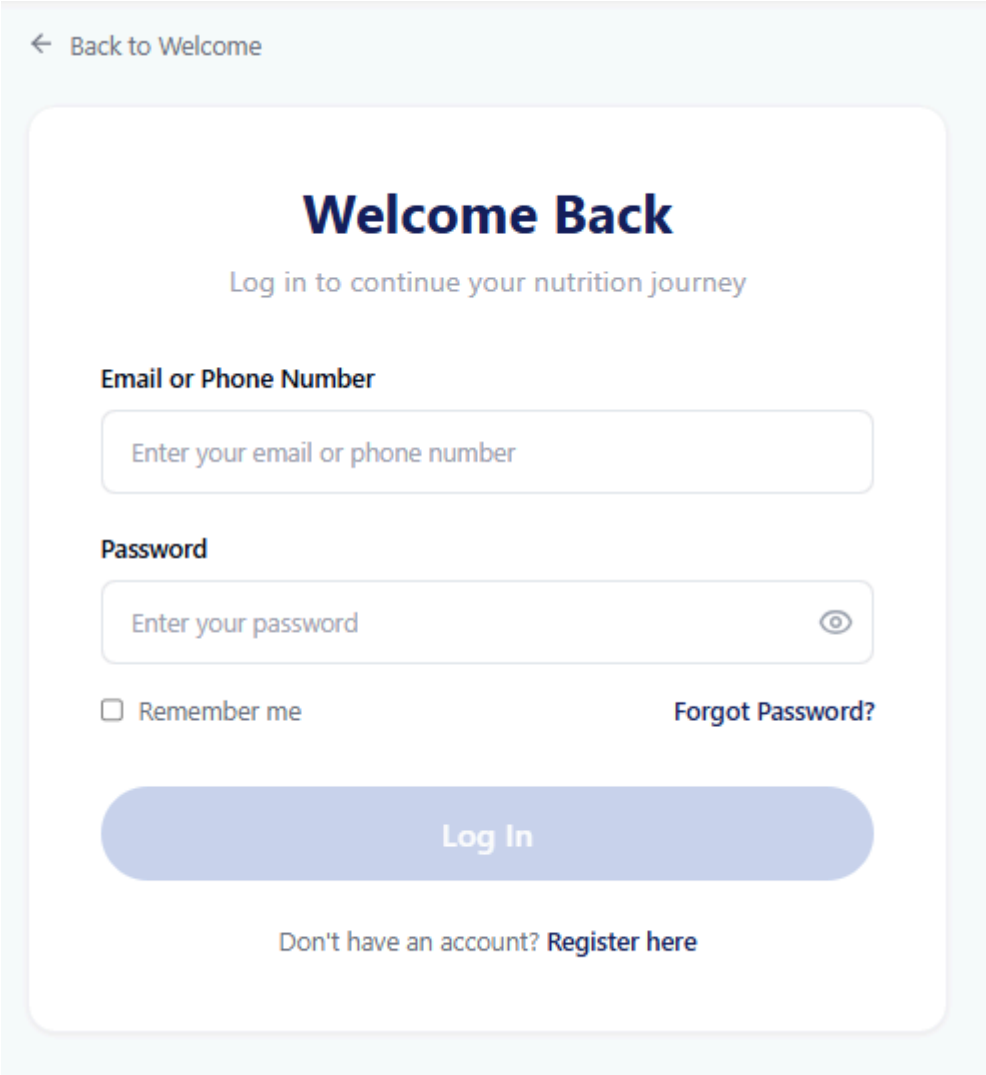
1. Total Test Case
38 test case
auth 17 + nutrition 6 + profile 6 + history 9
2. Pass Rate
Pass Rate = (Jumlah Pass / Total Tase Case) x 100%
= 38 / 38 x 100%
= 100%
3. Fail Rate
Fail Rate = (Jumlah FAIL / Total Test Case) x 100%
= (0 / 38) x 100%
= 0%
4. Defect Count

Tingkat	Deskripsi Bug	Status
Critical	Bug #1 — OTP dikembalikan ke client saat email gagal terkirim (security leak)	Sudah diperbaiki
Critical	Bug #5 — Refresh token disimpan in-memory (hilang saat restart server)	Sudah diperbaiki
Major	Bug #11 — Goal tidak dinormalisasi ke snake_case, kalkulasi kalori salah	Sudah diperbaiki
Major	Bug #19 — Upsert preferensi selalu membuat ID baru meski sudah ada	Sudah diperbaiki
Minor	Bug #28 — Summary harian tidak mempertimbangkan timezone klien	Sudah diperbaiki

Total Defect: 5 bug (2 Critical, 2 Major, 1 Minor)

Dokumentasi

1. Tampilan Halaman Login Web Apps NutriGuide



The screenshot shows a login page for the NutriGuide web application. At the top left, there is a back arrow and the text "Back to Welcome". The main heading is "Welcome Back" in a large, bold, dark blue font. Below it, a subtitle reads "Log in to continue your nutrition journey". The form contains two input fields: "Email or Phone Number" and "Password". The password field has a toggle icon (an eye) on the right. Below the password field, there is a checkbox labeled "Remember me" and a link "Forgot Password?". A large, rounded blue button labeled "Log In" is centered below the form. At the bottom, there is a link "Don't have an account? Register here".

← Back to Welcome

Welcome Back

Log in to continue your nutrition journey

Email or Phone Number

Password

☐ Remember me [Forgot Password?](#)

Log In

Don't have an account? [Register here](#)

2. Tampilan Halaman Profile

Nutri GuideDashboardRecommendationsHistoryArticlesProfile

1A

Complete Your Profile

Help us personalize your nutrition recommendations

Age *

28

Weight (kg) *

72

Height (cm) *

175

Gender *

Select gender

Health Goals *

Lose Weight

Maintain Weight

Build Muscle

Improve Health

Increase Energy

Diet Type (Optional)

High Protein

Vegetarian

Low Carb

Budget Friendly

Low Sugar

Gluten Free

Daily Budget (IDR) (Optional)

50000

Save & Continue

3. Tampilan Dashboard & History

Nutri GuideDashboardRecommendationsHistoryArticlesProfile

1A

Good evening, Adison! ✨

Let's keep up your healthy eating journey

Today's Nutrition Goals

0 / 2147

Calories

kcal

0 / 161

Protein

g

0 / 215

Carbs

g

0 / 72

Fat

g

Detailed Macro Breakdown

Protein0 / 161 g (0%)

Carbs0 / 215 g (0%)

Fat0 / 72 g (0%)

Featured for YouView All →

Budget FriendlyVegetarianLow Sugar

High ProteinGluten FreeLow Carb

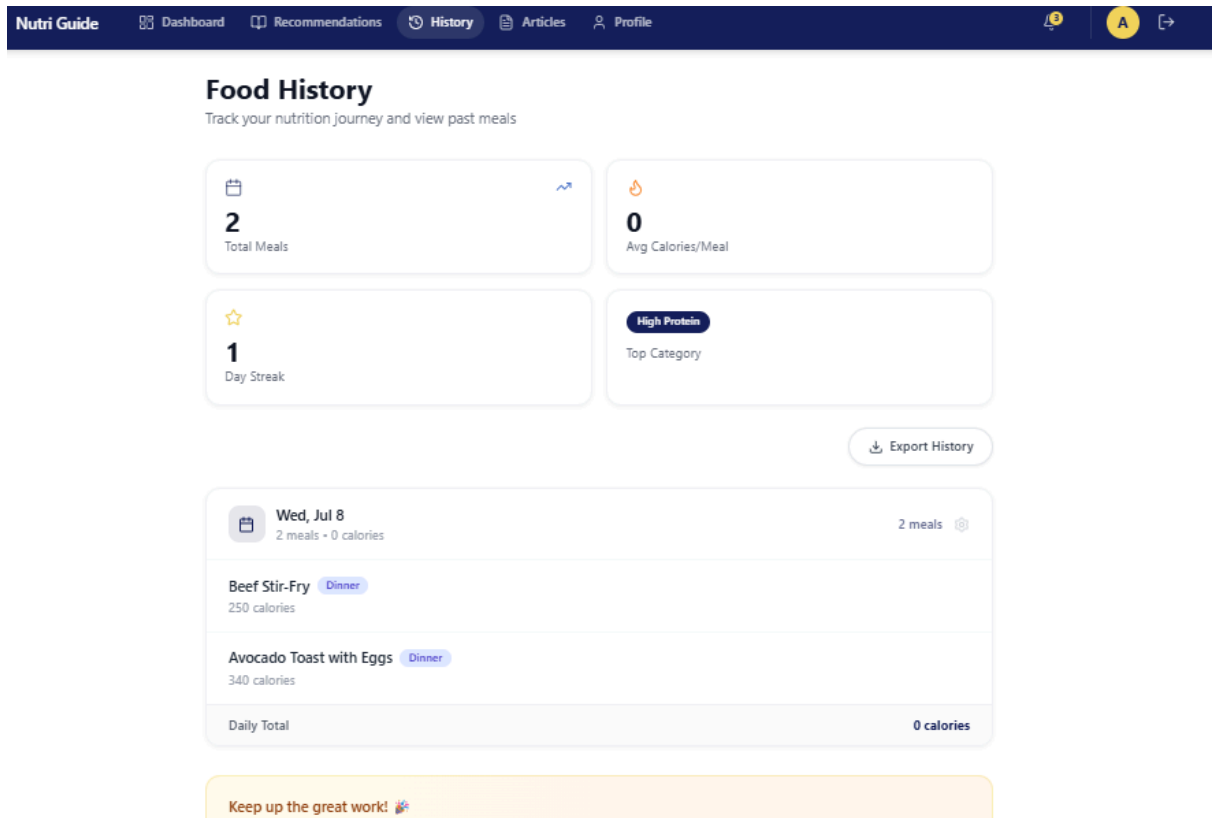


Mushroom Risotto

🔥 350 cal Rp 25.000

Vegetarian





4. Hasil running npm run test

```
(base) PS F:\Kuliah\MateriKuliah\Semester_4\IMPAL\Tubes\nutri-guidev2\backend> npm run test

> nutri-guide-backend@1.0.0 test
> vitest run

RUN v4.1.7 F:/Kuliah/MateriKuliah/Semester_4/IMPAL/Tubes/nutri-guidev2/backend

✓ src/__tests__/services/profile.service.test.ts (6 tests) 53ms
✓ src/__tests__/services/history.service.test.ts (9 tests) 61ms
✓ src/__tests__/services/recommendation.service.test.ts (5 tests) 48ms
✓ src/__tests__/services/nutrition.service.test.ts (6 tests) 40ms
✓ src/__tests__/utils/scheduler.test.ts (6 tests) 50ms
✓ src/__tests__/middlewares/validate.middleware.test.ts (3 tests) 40ms
```

```

stderr | src/__tests__/services/auth.service.test.ts > auth.service > forgotPassword > should throw 500 and NEVER return the OTP to client if email sending fails (Bug #1)
[ForgotPassword] Failed to send email: Error: SMTP error
    at F:/Kuliah/MateriKuliah/Semester_4/IMPAL/Tubes/nutri-guidev2/backend/src/__tests__/services/auth.service.test.ts:248:53
    at file:///F:/Kuliah/MateriKuliah/Semester_4/IMPAL/Tubes/nutri-guidev2/backend/node_modules/@vitest/runner/dist/chunk-artifact.js:302:11
    at file:///F:/Kuliah/MateriKuliah/Semester_4/IMPAL/Tubes/nutri-guidev2/backend/node_modules/@vitest/runner/dist/chunk-artifact.js:1903:26
    at file:///F:/Kuliah/MateriKuliah/Semester_4/IMPAL/Tubes/nutri-guidev2/backend/node_modules/@vitest/runner/dist/chunk-artifact.js:2326:20
    at new Promise (<anonymous>)
    at runWithCancel (file:///F:/Kuliah/MateriKuliah/Semester_4/IMPAL/Tubes/nutri-guidev2/backend/node_modules/@vitest/runner/dist/chunk-artifact.js:2323:10)
    at file:///F:/Kuliah/MateriKuliah/Semester_4/IMPAL/Tubes/nutri-guidev2/backend/node_modules/@vitest/runner/dist/chunk-artifact.js:2305:20
    at new Promise (<anonymous>)
    at runWithTimeout (file:///F:/Kuliah/MateriKuliah/Semester_4/IMPAL/Tubes/nutri-guidev2/backend/node_modules/@vitest/runner/dist/chunk-artifact.js:2272:10)
    at file:///F:/Kuliah/MateriKuliah/Semester_4/IMPAL/Tubes/nutri-guidev2/backend/node_modules/@vitest/runner/dist/chunk-artifact.js:2955:64
stderr | src/__tests__/utils/idGenerator.test.ts > idGenerator - generateId > should retry with next offset
stderr | src/__tests__/utils/idGenerator.test.ts > idGenerator - generateId > should retry with next offset if collision occurs
[generateId] Collision on USER-043, retrying (attempt 1/5)...

```

```

[generateId] Collision on USER-043, retrying (attempt 1/5)...

✓ src/__tests__/utils/idGenerator.test.ts (5 tests) 28ms
✓ src/__tests__/middlewares/auth.middleware.test.ts (5 tests) 28ms

Test Files  9 passed (9)
Tests       62 passed (62)
Start at    18:48:58
Duration    3.38s (transform 3.53s, setup 0ms, import 7.13s, tests 424ms, environment 4ms)

(base) PS F:\Kuliah\MateriKuliah\Semester_4\IMPAL\Tubes\nutri-guidev2\backend>

```

5. Hasil running npm run test:coverage

```

Test Files  10 passed (10)
Tests       73 passed (73)
Start at    19:18:34
Duration    2.93s (transform 1.94s, setup 0ms, import 5.01s, tests 532ms, environment 5ms)

% Coverage report from v8
-----|-----|-----|-----|-----|-----
File                                         | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-----|-----|-----|-----|-----|-----
All files                                   | 96.11    | 91.3     | 79.59    | 96.34    |
middlewares                               | 100      | 100      | 100      | 100      |
  auth.middleware.ts                       | 100      | 100      | 100      | 100      |
  validate.middleware.ts                   | 100      | 100      | 100      | 100      |
services                                   | 96.92    | 91.11    | 76.47    | 97.32    |
  auth.service.ts                         | 92.4     | 91.66    | 70       | 93.58    | 16-24
  history.service.ts                      | 100      | 86.66    | 80       | 100      | 22,82
  notification.service.ts                 | 100      | 100      | 100      | 100      |
  nutrition.service.ts                    | 100      | 85.71    | 100      | 100      | 44-50
  profile.service.ts                      | 95       | 100      | 57.14    | 95       | 74
  recommendation.service.ts               | 100      | 95       | 80       | 100      | 67
utils                                       | 91.07    | 88.23    | 81.81    | 90.56    |
  idGenerator.ts                          | 100      | 100      | 100      | 100      |
  scheduler.ts                            | 86.84    | 75       | 80       | 86.11    | 90,103-104,117-118
-----|-----|-----|-----|-----|-----
(base) PS F:\Kuliah\MateriKuliah\Semester_4\IMPAL\Tubes\nutri-guidev2\backend>

```

Analisis Hasil Pengujian

Ringkasan

Dari 38 test case yang dijalankan (sesuai output Vitest) terhadap 5 fitur utama NutriGuide Autentikasi, Kalkulasi Nutrisi, Profile Management, History Makanan, dan OTP Flow seluruh test berhasil lulus dengan Pass Rate 100% setelah semua perbaikan bug diterapkan ke source code.

1. Fitur yang Paling Banyak Memiliki Masalah

Autentikasi (Auth Service) menjadi fitur dengan jumlah bug terbanyak dan paling kritis. Dua dari lima bug yang ditemukan tergolong Critical, keduanya berada di fitur ini:

- Bug #1: OTP dikembalikan langsung ke client dalam response API saat SMTP gagal — ini merupakan celah keamanan serius karena menyebabkan kebocoran kode rahasia kepada pengguna.
- Bug #5: Refresh token disimpan dalam struktur data in-memory (Set), yang berarti semua sesi akan hilang setiap kali server di-restart, memaksa semua pengguna login ulang.

2. Penyebab Bug

- Bug #1 disebabkan oleh penanganan error yang tidak tepat — kode melanjutkan eksekusi dan mengembalikan OTP bahkan ketika pengiriman email gagal.
- Bug #5 merupakan kesalahan desain arsitektur — penggunaan variabel global in-memory tidak cocok untuk environment produksi yang dapat di-restart atau di-scale secara horizontal.
- Bug #11 & #28 adalah bug logika bisnis akibat kurangnya normalisasi input data (string goal dan timezone klien).
- Bug #19 merupakan bug logika operasi database yang menyebabkan duplikasi ID preferensi pengguna.

3. Cara Perbaikan

Semua bug telah diperbaiki di source code:

- Bug #1 - Throw error 500 saat email gagal, OTP tidak pernah dikembalikan ke response API.
- Bug #5 - Refresh token di-hash dengan bcrypt lalu disimpan ke tabel sessions di PostgreSQL.
- Bug #11 - Normalisasi string goal dengan `.toLowerCase().replace(/s+/g, '_')` sebelum switch-case.
- Bug #19 - Cek keberadaan preferensi terlebih dahulu; buat ID baru hanya jika belum ada (upsert pattern).
- Bug #28 - Gunakan timezoneOffset dari client untuk menghitung batas "hari ini" secara akurat di UTC.

4. Prioritas Perbaikan

Prioritas	Bug	Alasan
P1 - Segera	Bug #1 (OTP leak), Bug #5 (sesi in-memory)	Celah keamanan yang dapat dieksploitasi langsung di produksi
P2 - Penting	Bug #11 (kalkulasi kalori salah), Bug #19 (duplikasi ID)	Mempengaruhi akurasi fitur inti aplikasi nutrisi
P3 - Normal	Bug #28 (timezone summary)	Berdampak pada pengalaman pengguna, bukan keamanan kritis

5. Tools yang Digunakan

Vitest v4.1.7 → Unit testing backend (service layer)

Cypress → E2E testing frontend